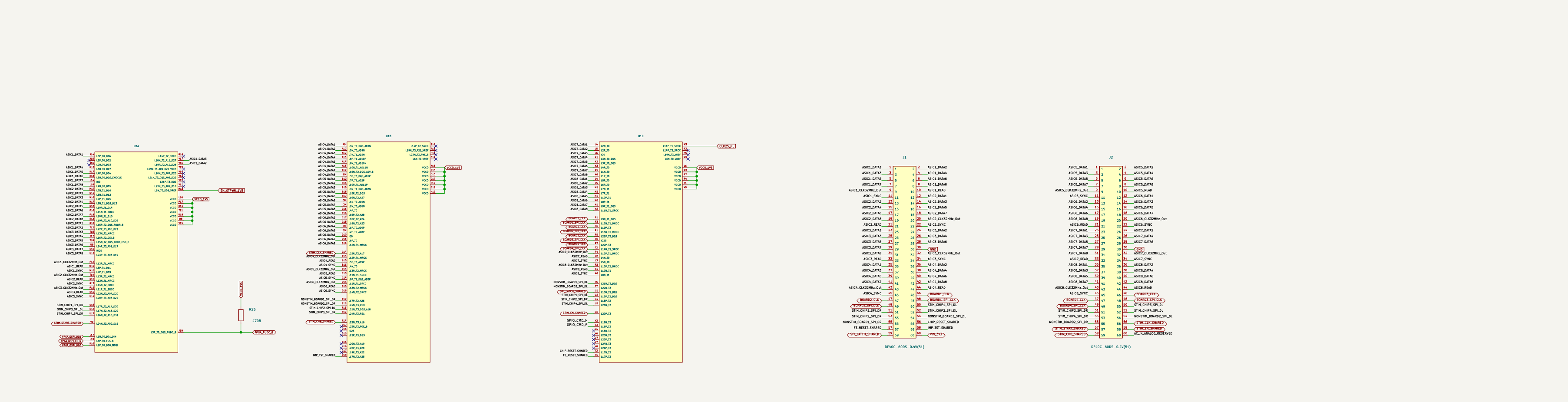
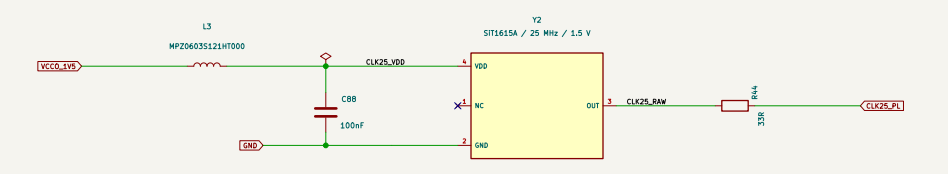
01. FPGA I/O + ASIC INTERFACES + 25 MHz CLOCK | Trenz p5

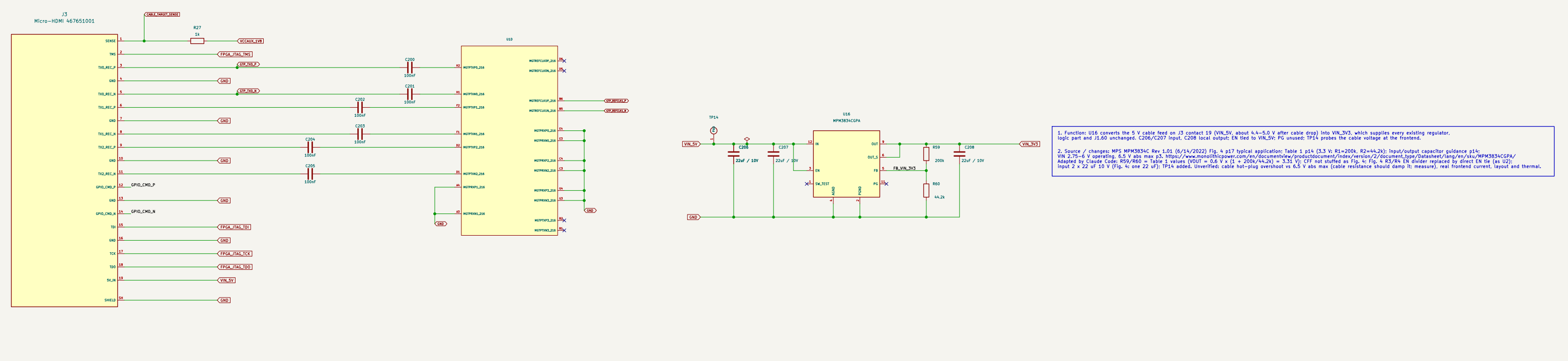


1. Function: U1 / J1 / J2: Route116 active signals for eight1.5V ASICs through two60-contact mezzanines. J1.60 carries U16-regulated VIN_3V3; J2.60 remains reserved analog.
2. Source / changes: Codex custom ASIC interface; inherited V54/V62 pin contract preserved. ASIC_REFERENCE.md sections2-5; Chip_FPGA Interface.pptx slides13-17/25-28; Pins.xlsx. Trenz p5 I/O bank use adapted to ASIC allocation. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF ASIC silicon timing/rail limits remain unqualified.

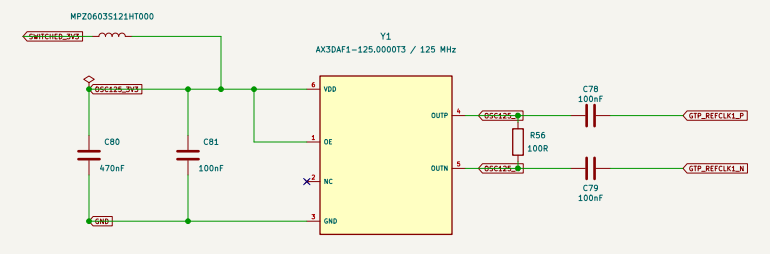


1. Function: Y2:25MHz fabric clock drives N3 SRCC at1.5V. L3 filters1.5V: C88=100nF; R44=33R output damping.
2. Source / changes: Trenz p5 U8/L2/C24/R23 structure adapted: SIT1615 1.5V replaces SIT8008 1.8V; T14 occupied by ASIC2 clock so useN3. C88 retained100n per SiTime; R44 exact Trenz RC0201FR-0733RL restored. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF https://www.sitime.com/datasheet/SiT1615 Timing/firmware constraints remain unverified.

02. GTP CHANNELS + 125 MHz REFERENCE | Trenz p6

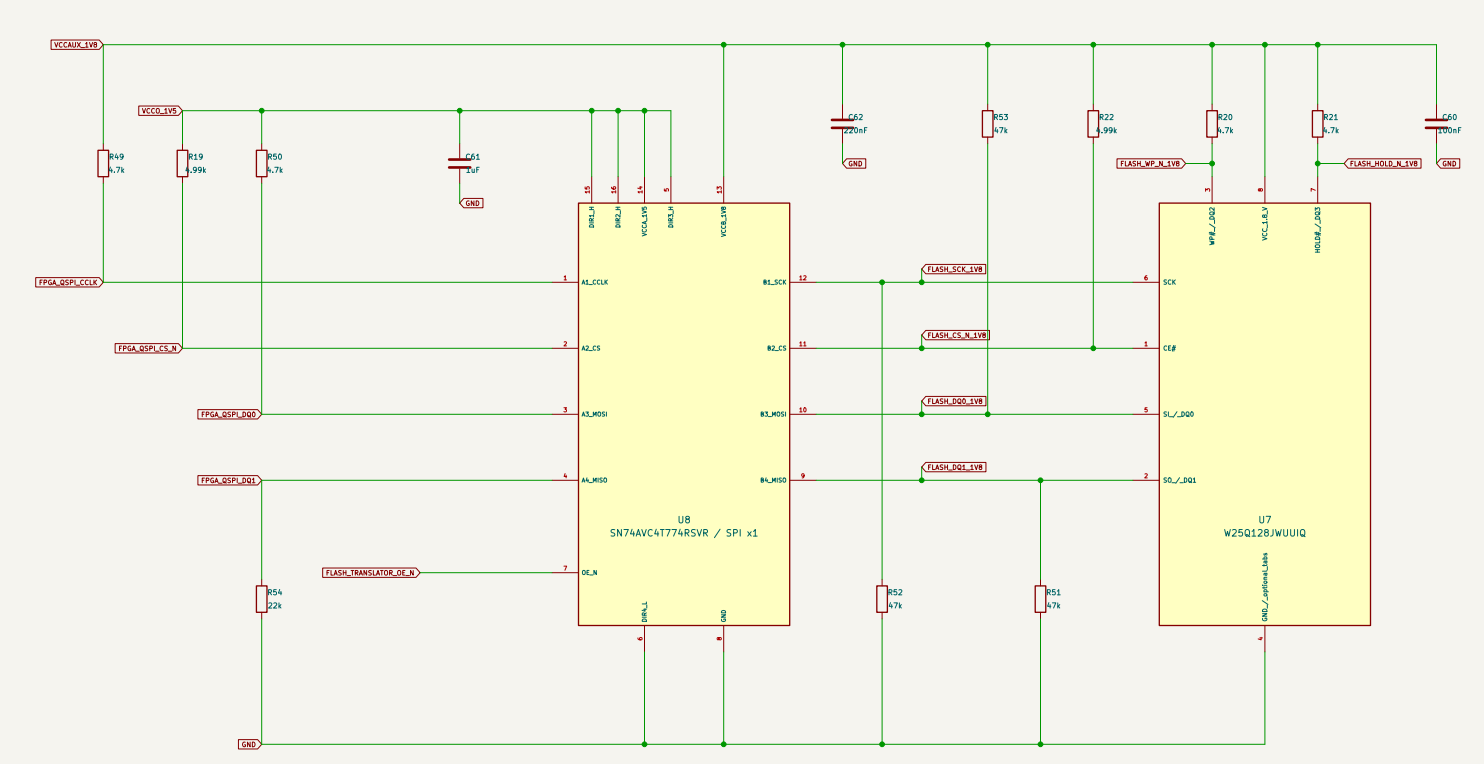


1. Function: J3 carries three GT recording TX pairs, regular-I/O commands, JTAG and 5 V power (contact 19 -> VIN_5V -> U16). TX0 connects to XEM; TX1/TX2 end open at carrier.
2. Source / changes: Codex custom HDMI map adapted from V3; 9/11 add TX2 via C204/C205; 12/14 move from GTP RX1 to U1 V3/V2 regular I/O.
AMD DS181 v1.27.1 Table 50: CEXT=100 nF; C200-C205 are candidates. https://docs.amd.com/api/khub/documents/lAkxxTOk96ANLJqYf2hgrQ/content
Unused RX1 A4/A3 grounded: AMD UG482 v1.9 Table 5-6 pp.223-224. https://docs.amd.com/api/khub/documents/Xk_zMbXsJ92suJ4_0Akkg/content
1.5 V Bank34 command proposal: DIFF_SSTL15, IN_TERM=UNTUNED_SPLIT_50: AMD UG471 v1.10 Table 1-7. https://docs.amd.com/api/khub/documents/lbGcnPFe6eF19RHma_Y-lA/content
Carrier needs compatible regular I/O and VIO1=1.5 V. TX0 connects to XEM; TX1/TX2 are future carrier NC. Pin/clock/protocol, coupling, cable SI, shell bonding and unpowered-target protection remain unverified. V5 changes contact 19 from 3.3 V to 5 V (carrier TPS2553 port feed).



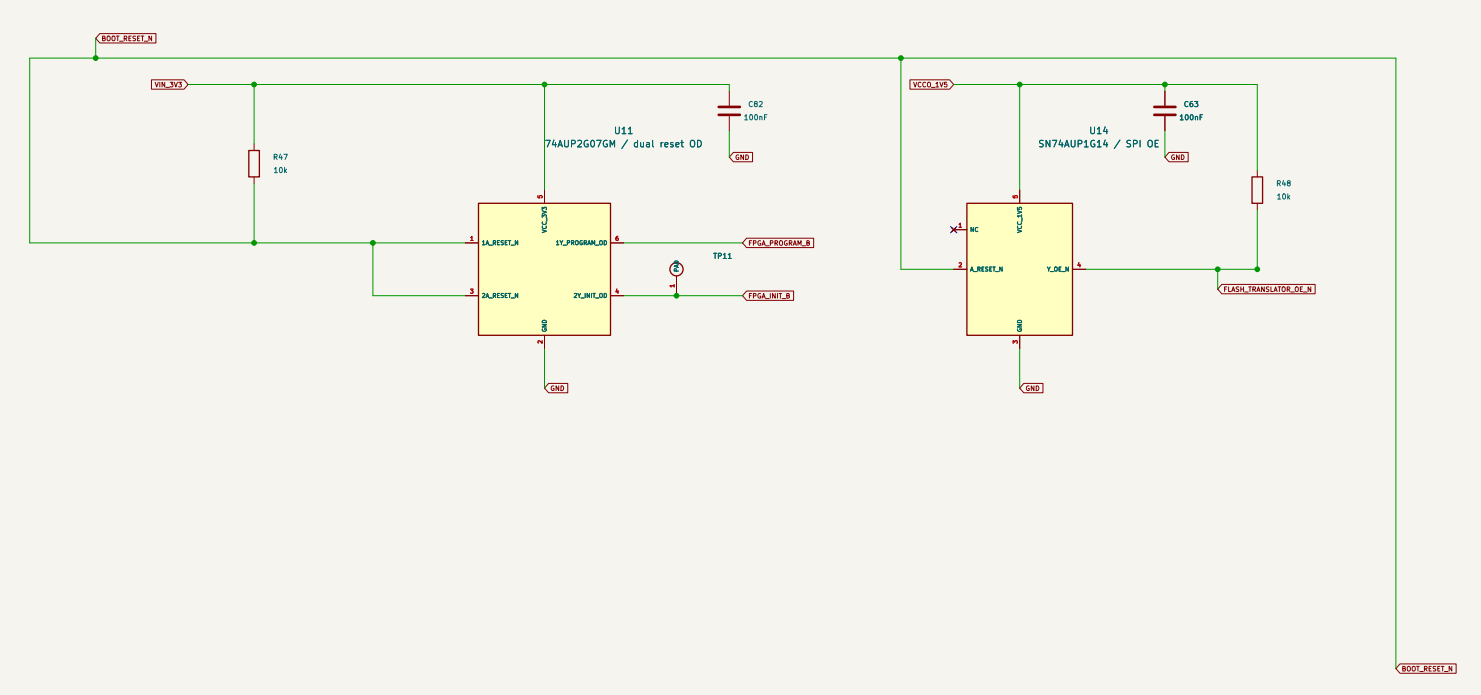
1. Function: Y1 supplies125MHz LVDS through C78/C79=100nF to REFCLK1P/N. L4 filters switched3.3V; C80=470nF and C81=100nF bypass it. R56=100R loads OSC125_P/N before the coupling capacitors; OE HIGH, pin2 NC.
2. Source / changes: Trenz TE0714-04-42I-7-L REV04 p6:125MHz LVDS U2, L1 ferrite, C10=470nF, R1=100R and C134/C135=100nF coupling. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Adapted by Codex: Y1=AX3DAF1-125.0000T3 (exact Trenz oscillator MPN undisclosed). C80=470nF, C81 10uF->100nF; added L4/R56. Abracon AX3 rev25Feb2022 pp3,9,12-13 requires active-HIGH OE and shows100nF local bypass. https://abracon.com/datasheets/AX3.pdf REFCLK1 B6/B5 matches Trenz; P-to-P/N-to-N retained. AMD UG482 v1.9 pp225-227,235: LVDS AC-coupled reference. https://docs.amd.com/v/u/en-US/ug482_7Series_GTP_Transceivers Clock jitter, filtered-rail noise, layout and hardware startup remain unqualified. R56 uses the same100R0.1%25ppm candidate as R16; exact source MPN is undisclosed.

03. FLASH + SPI TRANSLATION | Trenz p6 adaptation

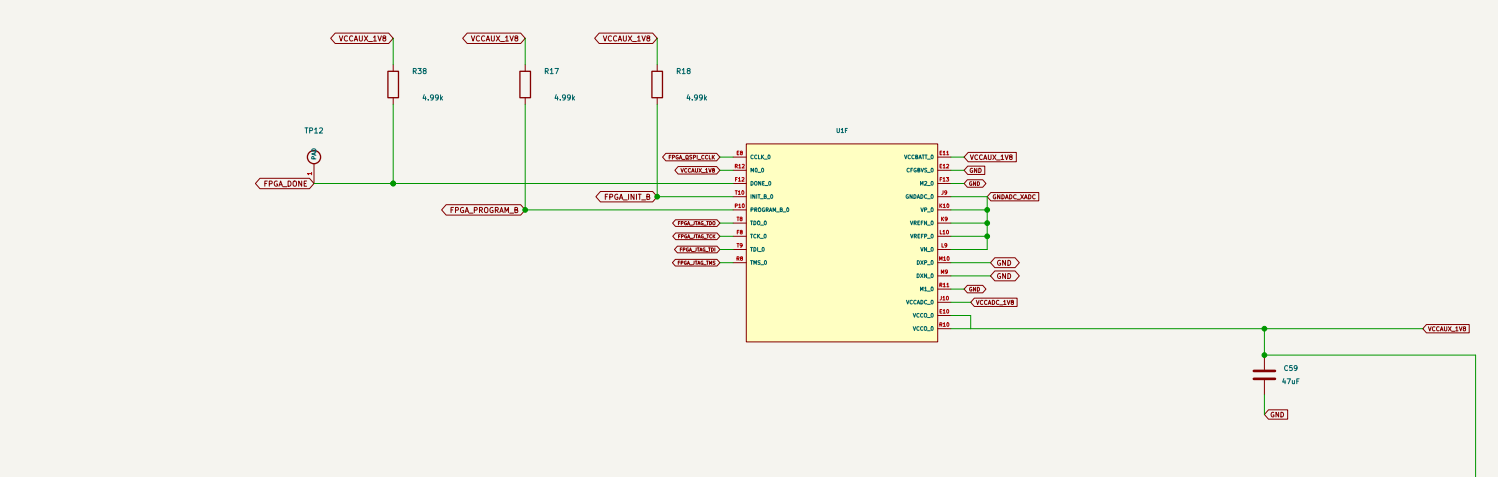


1. Function: U7/U8: 16MiB 1.8V SPI NOR flash (Winbond W25Q128JWUUIQ, 4x3mm USON-8) and SN74AVC4T774RSV SPIx1 translation. Bank14 remains1.5V; initialCONFIGRATE3MHz/QE0/3-byte addresses.
2. Source / changes: Adapted from V5 IS25WP128F-JKLE (6x5mm WSON-8) to Winbond W25Q128JWUUIQ (4x3mm USON-8, pkg code UU, 0.80mm pitch; AMD UG908 w25q128fw/jw family) to save 60% flash area for V6 size reduction. R19/R22 4.99k; pullups 1.8V. https://www.winbond.com/resource-files/w25q128jw%20revg%2008242021%20plus.pdf https://www.ti.com/lit/ds/symlink/sn74avc4t774.pdf Land pattern & boot timing unverified.

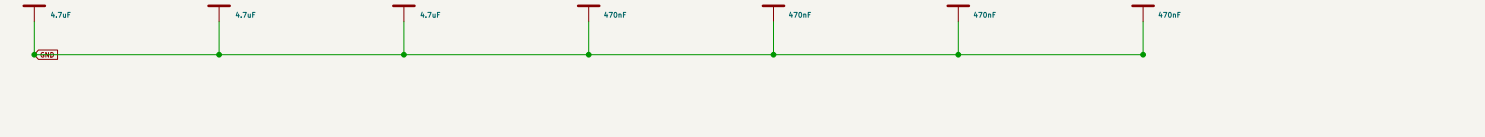
04. CONFIGURATION + RESET OUTPUTS | Trenz p7 roles



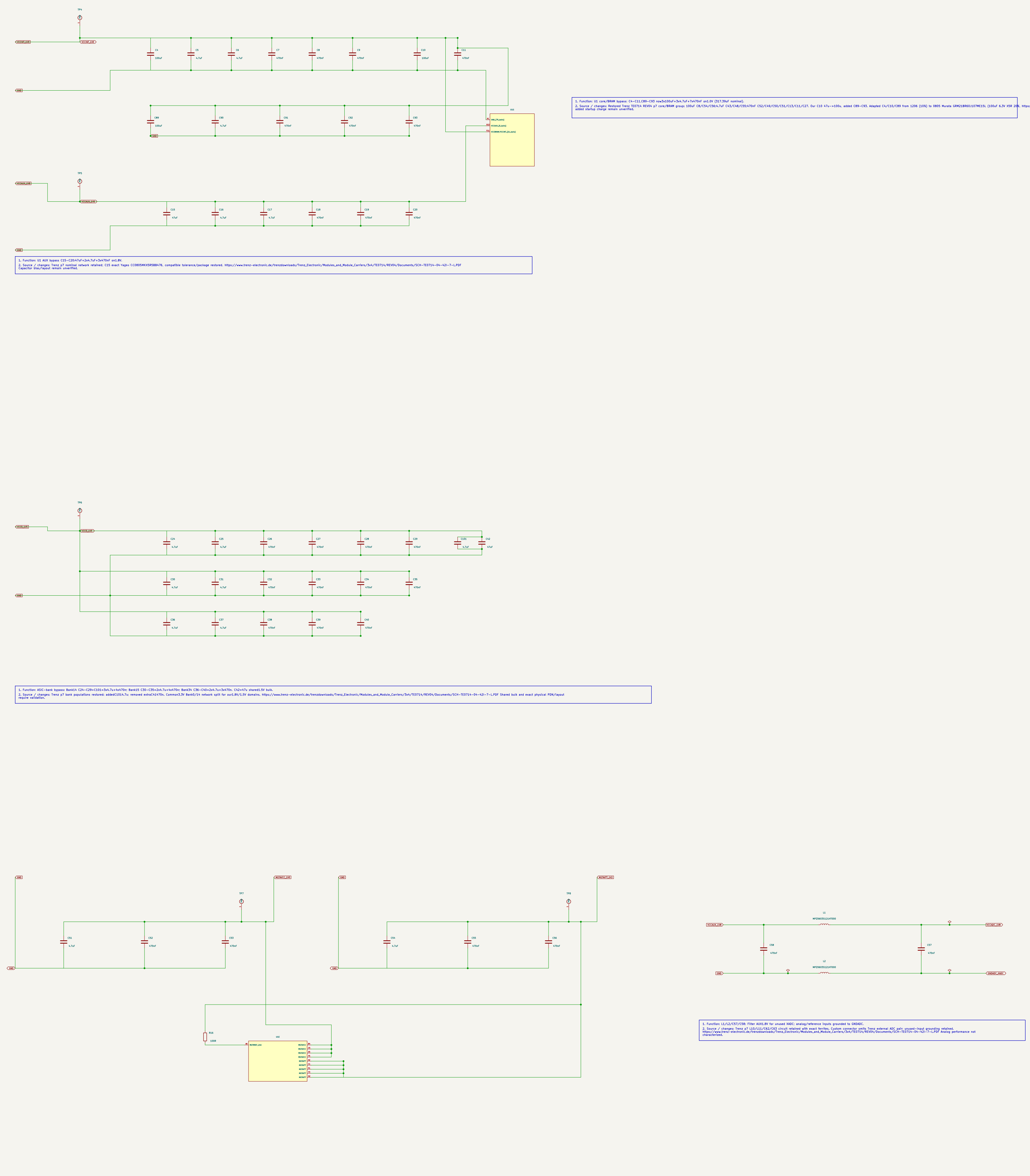
1. Function: U11 holds PROGRAM_B and INIT_B LOW through two independent open-drain outputs; R17/R18 release them to 1.8 V. U14 drives the 1.5 V active-low U8 enable. C82/C63 are local bypass.
2. Source / changes: Correspondence: Trenz p7 D1/D2 reset-hold functions -> U11; U14 is added for our U8 flash translator, not Trenz boot-mode inverter U9. V64 electrical network unchanged. Codex voltage adaptation of Trenz p7 D1/D2 reset function; inherited V61 agreed flash/U8 support retained. Trenz diode/common3.3V solution is not used across our1.8V/1.5V domains. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF https://assets.nexperia.com/documents/data-sheet/74AUP2G07.pdf https://www.ti.com/lit/ds/symlink/sn74aup1g14.pdf



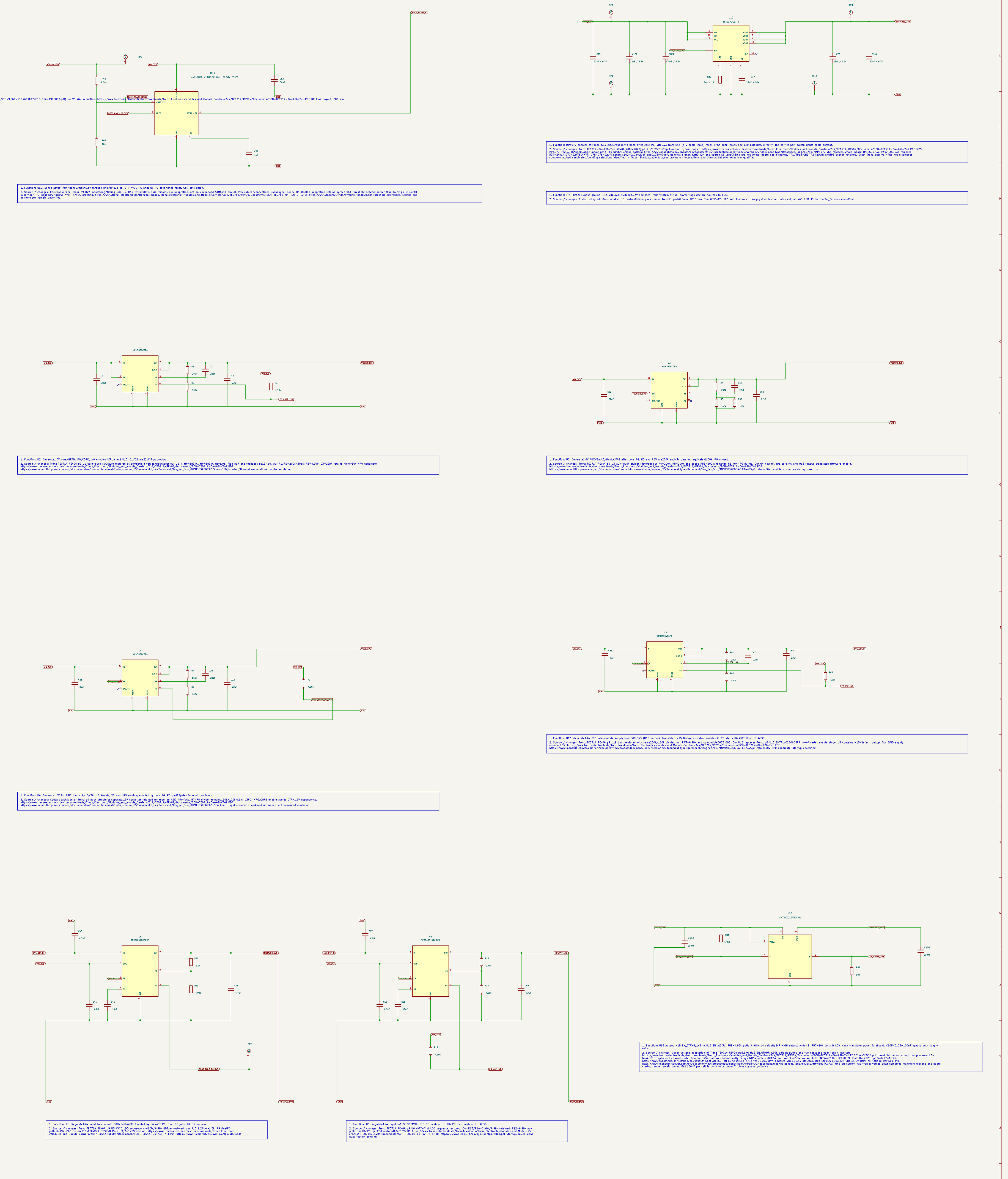
1. Function: U1 configuration: Master SPI001,CFGBVS GND, Bank0/JTAG1.8V; ASIC banks1.5V. PROGRAM_B and INIT_B reset independently. VCCBATT E11 now AUX1.8V.
2. Source / changes: Trenz p7 values restored where compatible: R17/R18/R19/R22=4.99k; E11 GND->AUX. Mode ties/1.8V rails and x1 flash remain agreed adaptations.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF AMD UG470v1.17 Table2-4 permits AUX for unused BBRAM.
https://docs.amd.com/api/khub/documents/FOs3lXmlcWxBhTlFxVKyGA/content



05. FPGA SUPPLY PINS + BYPASS + XADC | Trenz p7



06. REGULATORS + READINESS SUPERVISOR | Trenz p9 roles



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Title: FPGA V54 dual60 direct FPGA/LDO interface – unrouted

Size: User | Date: 2026-10-19 | **Rev: V54**
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